

YUVARAJ MUNIYANDI

AI Solutions Architect | Technology Delivery Leadership | Enterprise Data & AI Platforms

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PROFESSIONAL SUMMARY

Technology leader with 18+ years of experience spanning enterprise delivery leadership and hands-on AI/data platform architecture, with \$16M+ in measurable business impact across six production AI and data systems in banking, healthcare and fintech. Currently AI Solutions Architect at Exasol, personally designing and building data-federation architecture. Previously AI Architect at Maveric Systems, where I architected three enterprise AI platforms (vendor intelligence, document intelligence, invoice automation) across a 30-person team. Before that, 15+ years leading AI/ML and technology delivery programs at EXL, Crayon Data, Applied Data Finance and Accenture, coordinating architecture decisions with business outcomes and directing teams of up to 60 people. I build hands-on to stay technically credible: I independently designed and built NexusIQ, a working multi-agent AI governance platform, and NexusRetail, a production-shaped multi-tenant SaaS system on AWS built end-to-end while preparing for the AWS Solutions Architect Associate certification.

CORE EXPERTISE

- Enterprise AI Architecture: multi-agent systems, RAG, semantic document search, knowledge graphs, LLM orchestration, AI governance/RBAC, data federation (Exasol Virtual Schema)
- Cloud & Infrastructure: AWS (VPC, ECS Fargate, RDS, Cognito, S3/CloudFront, Route 53, ACM), Azure, GCP, Terraform, Docker, Kubernetes (working knowledge)
- Modern Data Platforms: Exasol, Snowflake, Databricks, federated/virtual-schema query architectures, MPP analytics
- Technology Delivery Leadership: AI/ML product delivery, program management, cross-functional team leadership of up to 60 people, stakeholder and vendor management
- AI/ML Frameworks & Tooling: LangChain, LangSmith, OpenAI, Gemini, Neo4j, Document AI, Texttract, FlanT5, Scikit-Learn (decision trees, gradient boosting)
- MLOps & Production Reliability: model monitoring, A/B testing frameworks, LLM observability and tracing (LangSmith), release/deployment management, CI/CD, containerization (Docker, Kubernetes), Infrastructure as Code (Terraform)

PROFESSIONAL EXPERIENCE

Solution AI Architect, Exasol

April 2026 to Present

- Personally designed and built a Virtual Schema integration connecting Exasol with Snowflake for federated queries. This avoided a separate data-migration pipeline and eliminated data-duplication and leakage risk.
- Independently created and delivered technical demo content, presenting Exasol's architecture and federation capabilities at customer sessions and public technical events including AI Summit Chennai, meetups and webinars. This contributed to an 8% increase in platform user engagement.

AI Architect, Maveric Systems

May 2025 to April 2026

- Architected the Vendor Intelligence Automation platform: a RAG-based chatbot with SharePoint ingestion, vector embedding, and a graph retrieval engine with an intelligent router for hybrid structured and semantic queries, plus event-driven multi-channel automation via Power Automate. Cut decision time by 85%, generating \$3M+ in annual savings. Stack: Python, Azure, React, LangChain.
- Designed a secure, multi-tenant Document Intelligence Platform on Google Cloud with a Neo4j knowledge graph for relationship mapping across the document corpus, improving onboarding speed and compliance across geographies and cutting onboarding time by 60%. Stack: Python, LangChain, OpenAI, Document AI.
- Built a multi-cloud Invoice Intelligence Platform: secure authentication via Workload Identity Federation and the Graph API, Document AI processing with a business-rules engine, and automated sync to BigQuery, replacing a fully manual multi-day process. Lifted accuracy by 40%, cut manual effort by 70%, and saved \$1.5M annually. Stack: Python, GCP, React, Gemini Flash.
- Designed a governance-first data-access pattern for the Invoice Intelligence Platform: rather than exposing raw invoice records to the LLM, an MCP tool exposes only table schema/metadata, letting the model generate a SQL query that the database executes directly — the LLM only ever sees query results, never raw row-level data.

- Advised on a separate team's client engagement where internal infrastructure policy (GCP-only provisioning) conflicted with the client's requirement to keep data in AWS S3. Designed and demoed a proof-of-concept using GCP Vertex AI for inference while data remained in place in S3, accessed via WAF, resolving the conflict without data migration and securing client sign-off.
- Performed cloud cost estimation and trade-off analysis across projects, presenting evidence-based cost justifications to align proposed architecture scope with available budget and secure funding approval.
- Defined the cloud-native deployment standard, including containerization, Kubernetes, Infrastructure as Code and CI/CD, across all three platforms.
- Directed a 30-member architecture, engineering and data-science team spanning all three platforms.

Senior Manager, AI/ML Product Delivery, EXL Services, Healthcare Payment Analytics

October 2023 to May 2025

- Led product delivery and program management for healthcare AI/ML solutions, coordinating technical architecture with business objectives.
- Managed delivery of a GenAI-powered rule-creation and anomaly-detection platform, collaborating with architects on system design for claims automation. Coordinated implementation of LLM-based text analysis (FlanT5), AWS Textract document parsing, a custom rule engine translating natural-language policy into executable business rules, and the analytics pipeline.
- The program delivered a 90% reduction in cycle time and a 70% improvement in efficiency for policy logic creation and medical reviews.
- Drove \$11M in revenue through strategic cloud migration and vendor consolidation initiatives.

Senior Program Manager, Crayon Data, Fintech SaaS (Recommendation Engine)

April 2021 to October 2023

- Managed delivery of enterprise SaaS implementations for banking clients, coordinating technical architecture with business requirements.
- Delivered 7+ high-value banking client implementations, managing integration of the recommendation engine with client data infrastructure. Coordinated technical teams on architecture decisions covering API design, data governance, model orchestration (decision trees, collaborative filtering) and deployment patterns. Stack: Python, AWS.
- Generated AED 2.3M in incremental revenue and AED 1.7M in affiliate revenue through optimized integration strategies, and improved client satisfaction by 30% through technical delivery excellence and proactive architecture guidance.

Senior Manager, Product Delivery, Applied Data Finance, B2C Fintech

August 2016 to April 2021

- Managed delivery of an ML-powered lending platform at scale, coordinating cross-functional teams of 60+ people across engineering, data science and operations.
- Led the delivery program for a machine-learning underwriting platform that reduced loan default rates by 20%, coordinating with data science on model architecture including decision trees, gradient boosting, logistic regression and feature-engineering pipelines. Stack: Python, Scikit-Learn.
- Managed the program to ensure production stability through model monitoring and an A/B-testing framework, reducing post-launch issues by 35% and improving feature engagement by 25%. Implemented deployment coordination and quality gates.

Delivery Lead / Project Manager, Accenture Services Pvt Ltd

April 2007 to January 2016

- Delivered major enterprise transformation programs for US-based clients across analytics, data platforms and application modernization, coordinating technical execution with business objectives across complex multi-workstream projects.

INDEPENDENT ARCHITECTURE PROJECTS

NexusIQ, Multi-Agent AI Governance Platform (yuvarajai.com/nexusiq)

Independently designed and built a working enterprise AI reference platform to test and demonstrate architecture patterns for federated data access, retrieval-augmented generation and AI governance. Used for ongoing hands-on research, technical demos and public speaking. This is a solo project, unrelated to employer IP.

- Multi-database federation via Exasol Virtual Schema: natural-language (Text-to-SQL) querying joined live across Exasol, Snowflake, Databricks, PostgreSQL and MySQL, without building a data pipeline.
- RAG over unstructured content: SharePoint document retrieval via Model Context Protocol (MCP), with Microsoft Entra ID SSO/OAuth and ChromaDB vector search.
- Governance layer: role- and department-based RBAC policy enforcement (approve, block, restrict) with a full audit trail, demonstrated live across an Analyst/Manager/Admin permission matrix.

- Six-layer agentic architecture: AI Gateway, Agent Planner, Knowledge Layer, Agent Memory, Enterprise Tools and Model Router.
- Bulk and single-file data ingestion (CSV/Excel to instantly queryable tables), plus a connection-management console supporting five database engines.

Stack: FastAPI, ChromaDB, MCP, Exasol, Snowflake, Databricks, MySQL, PostgreSQL, Microsoft Entra ID OAuth

NexusRetail, Multi-Tenant SaaS Platform on AWS (github.com/yuviim/nexusretail)

Independently designed and built a production-shaped multi-tenant inventory and order-management SaaS platform end to end on AWS, hands-on, block by block, while preparing for the AWS Solutions Architect Associate certification.

- Architected and implemented network isolation (VPC with public/private subnet segmentation and a NAT Gateway), a containerized API on ECS Fargate behind an Application Load Balancer, and RDS PostgreSQL with Secrets Manager-managed, rotatable credentials.
- Built Cognito authentication with per-tenant data isolation and a separate cross-tenant Super Admin authorization tier using Cognito Groups. Verified isolation live across two independently seeded tenant organizations.
- Deployed the React frontend via S3 and CloudFront with a custom domain and ACM-issued HTTPS certificates. The entire infrastructure is defined and version-controlled in Terraform.
- Debugged and documented real production incidents, including cross-architecture container build failures, IAM/Secrets Manager misconfigurations, CORS policy gaps and deployment-rollout failures, in a running technical reference log.

Stack: AWS (VPC, ECS Fargate, RDS, Cognito, S3, CloudFront, Route 53, ACM, Secrets Manager), Terraform, Node.js/TypeScript, Express, Prisma, React

PUBLICATIONS, VIDEO SERIES & SPEAKING

- Publishes ongoing architecture research and reference diagrams at yuvarajai.com covering Enterprise AI Systems, Modern Data Platforms and Sovereign AI Systems.
- Independently produces and hosts “Building Enterprise Agentic AI Systems” and “Engineering Sovereign AI Systems,” ongoing multi-episode YouTube technical series (youtube.com/@yuvarajplatforms) walking through enterprise agentic architecture layer by layer, from the AI Gateway through Model Router.
- Produces live architecture and demo walkthroughs of NexusIQ and Exasol internals, including Virtual Schema federation, in-memory processing, columnar storage and MPP execution, for a technical audience.
- Independently built and published a hands-on RAG and LLMOps learning project on LinkedIn: a LangChain-based mini RAG system with OpenAI embeddings, an in-memory vector store, a Streamlit interface, and a “Teacher Agent” explanation layer, with full prompt, cost, latency and retrieval tracing via LangSmith with measurable engagement on LinkedIn.
- Speaks at AI summits, enterprise architecture workshops and customer technical sessions on Enterprise AI Architecture and Modern Data Platforms.

EDUCATION & CERTIFICATIONS

- AWS Certified Solutions Architect, Associate (In Progress)
- Post Graduate Program in Artificial Intelligence, Texas McCombs School of Business
- Generative AI with Large Language Models, DeepLearning.AI
- Google Cloud Practitioner Fundamentals, Udemy
- SAFe 6.0, Scaled Agile Inc. (2023)
- Project Management Professional (PMP), PMI.org (2021)
- MBA, Operations Management, NIBM